# The empirical regulatory operator of $CD4^+$ T cells

Spectral denoising reveals modular, condition-gated and cell-type-specific perturbation structure

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## Abstract

Genome-scale Perturb-seq screens resolve thousands of regulators at once, but the resulting effects mix genuine signal with noise, connectivity hubs, and context dependence, and it is unclear whether they assemble into a reproducible regulatory architecture — or whether any such architecture generalizes beyond the perturbations observed. We address both questions in a genome-scale CRISPRi Perturb-seq atlas of primary human $CD4^+$ T cells by building a quality-audited regulatory operator — regulators filtered for reproducibility and context, effects taken as a single object rather than a hit list — and denoising it spectrally. After separating a global co-variation mode and a Marchenko–Pastur noise bulk, an empirically calibrated **92 signal directions** (not the 336 a closed-form edge would admit) carry a strongly non-random community structure (consensus modularity z = 259). Blind, annotation-free community detection recovers a known molecular complex: Mitochondrial Respiratory Chain Complex I, at BH-FDR $1.4 \times 10^{-7}$ — a stable, unsupervised community matching a curated complex, which validates the denoising, representation, and clustering at once. Convergent evidence — CORUM identity, CP-factor concentration, and cross-cell-type cohesion — supports a second, SAGA-centered coactivator module, and the module organization nominates autoimmune risk regulators against a length-matched null. The universal-versus-T-specific distinction the operator draws is sustained across three diverse cell types. Yet the same structure does not translate into prediction: real and permuted regulator features perform identically for leave-regulator-out inference, so unseen regulators are not predictable from available annotations. The perturbational operator contains recoverable biological structure — but recoverable structure is not equivalent to inductive predictability.

## 1. Introduction

Genome-scale Perturb-seq — pooled CRISPR perturbation with single-cell transcriptomic readout — has made it possible to measure the transcriptional consequence of knocking down essentially every expressed gene in a cell[1,2]. In primary human $CD4^+$ T cells, a recent genome-scale CRISPRi screen[3] extends this to a setting that matters for immunology and for therapy: four donors, three activation states (resting, early- and late-stimulated), and genome-scale coverage of candidate regulators. That screen — the dataset we analyze throughout — is the primary data source for every result below. The standard way such a screen is read is as a ranked list — which perturbations move the most genes, which are the "hub" regulators — and that reading is useful but incomplete. It treats each regulator's effect as an independent entry and discards the possibility that the effects share structure: that the response to one perturbation is informative about the response to another.

Here we take the opposite starting point. We treat the screen as a single object — the regulator × gene × condition tensor of perturbation effects, which we call the **regulatory operator** — and ask what its geometry reveals. This is a different question from ranking perturbations or extracting gene programs, and to our knowledge it has not been asked of a primary-cell Perturb-seq screen: existing analyses either rank regulators by effect breadth or factor the expression matrix into programs, but none builds the regulator × gene operator and interrogates it as an operator — its predictive structure, its low-dimensional subspace, its decomposition by condition. That is the gap this work addresses, and the results that follow are properties of the operator: it completes held-out conditions of observed regulators, it is gated by condition, its denoised structure resolves into reproducible modules — one of them a known molecular complex recovered blind to annotation — and it separates universal from cell-type-specific regulation, while failing to predict the trans-effects of regulators it never saw.

Positioning this against the single-cell-CRISPR methodology literature clarifies what is inherited and what is contributed. The front end — turning raw counts into calibrated, harmonized per-regulator effect estimates — is a solved problem we build on rather than redo: calibrated single-cell-CRISPR testing[4] and the scPerturb effect-quantification framework[5] define that layer. Our contribution sits above this calibrated front end: a spectrally denoised perturbational operator, its modular and condition-resolved organization, and an explicit separation between within-operator completion and prediction of unseen regulators. The closest methodological neighbor is guided sparse factor analysis[6], which recovers gene programs and their driving perturbations probabilistically; our CP/SVD operator decomposition is a deterministic analogue of that object, and we treat it as a comparison rather than a claim of priority. The predictive result is framed against a frontier where deep perturbation-response predictors do not yet reliably beat simple baselines[7,8]: we therefore ask not whether a flexible model can fit, but whether the operator's shared structure beats an honest persistence baseline out-of-panel — a lower, more falsifiable bar.

A methodological posture runs through all results, and we state it as a choice rather than leave it implicit: fewer claims, each controlled for the confound that most threatens it. Every result in this paper is gated by an explicit control — for the power–magnitude confound that inflates naive effect sizes, for cis-off-target artifacts, for donor reproducibility, and for decomposition stability — and where a control costs signal, we report the reduced number. In a field where predictive claims often do not survive a fair baseline, this discipline is part of the argument, not decoration around it.

A note on structure. The results open (Section 2, Figure 1) not with the operator but with the regulator ranking: the operator is built on top of the ranking, and the ranking is only trustworthy if its effect metric is not an artifact of statistical power — Section 2 establishes exactly that, so everything built on it stands on an audited foundation rather than an assumed one. On that substrate we (i) construct a quality-audited regulatory operator, filtering effects for reproducibility and context rather than trusting raw hubs or nominal significance; (ii) denoise it spectrally and show its community structure is strongly non-random; (iii) recover Mitochondrial Complex I by annotation-blind clustering, the cleanest evidence the operator contains real biological organization; (iv) identify a SAGA-centered coactivator module and condition-dependent programs by convergent evidence; (v) show the universal-versus-specific distinction is sustained across three cell types; and (vi) demonstrate, with a scrambled-feature control, that unseen regulators are not inductively predictable from the annotations available. The organizing conclusion is that a perturbational operator can contain recoverable biological structure while that structure remains predictively inaccessible from external covariates — a distinction we make explicit rather than eliding.

## 2. The regulator ranking is metric-robust and power-controlled

Before treating the screen as an operator, we establish that its most basic summary — which regulators have the broadest transcriptional effect — is trustworthy. This matters because the operator is built on the same effect estimates the ranking uses; if those estimates were dominated by a statistical-power artifact, the operator would inherit it. This section audits the ranking and shows it survives the three challenges most likely to undermine it: the choice of effect metric, the confound between apparent effect and assay power, and cis-off-target contamination. It is the foundation the rest of the paper stands on, not a fourth scientific claim.

### The ranking metric is not an artifact of the metric choice

A recurring criticism of Perturb-seq hub rankings is that they rank a *count* — how many genes pass a differential-expression threshold — rather than an effect size, so the ranking could be an artifact of the counting. We tested this directly by building a continuous effect magnitude (the summed absolute log-fold-change over the FDR-significant genes for each regulator) and comparing it to the count-based rank. The two agree closely: Spearman $\rho = 0.90$ between the proper FDR-restricted magnitude and `n_downstream` (per-gene $\rho = 0.93$). A naive all-gene magnitude — summing over every gene rather than the significant ones — is *worse*, not better, because it sums noise. The ranking is therefore robust to the metric: doing the effect-size calculation properly reproduces the count-based order, and the order does not depend on which of the two defensible metrics is used.

### The ranking is controlled for statistical power

The most serious confound is power. Regulators assayed in more cells can show larger apparent effects for purely statistical reasons, so a ranking that simply correlated with cell count would be measuring assay depth, not biology. In raw log-fold-change space this confound is real and large: the per-regulator effect-vector norm correlates with cell count at Spearman $\rho = -0.68$ (the sign reflects the particular normalization, but the magnitude is the concern). The ranking metric we use does not carry it: the FDR-restricted magnitude correlates with cell count at only $\rho = -0.21$, and `n_downstream` even less. The power confound is a property of the naive magnitude, not of the ranking, and it is precisely why every downstream operator analysis is performed in the pooled z-score space (Section 8) where the confound is removed.

Two further properties make the ranked signal interpretable. Effective knockdown gates it: 62% of contrasts have a statistically significant on-target knockdown, and those concentrate 85% of all trans-effects — so the broad-effect regulators are overwhelmingly the ones whose intended perturbation actually took hold. And the ranking is externally corroborated: the top regulators overlap an independent T-cell CRISPR screen at 25–30 of the top 100 ($p < 1e\text{-}26$), a concordance far above chance for two screens with different readouts.

### Donor reproducibility is reported, not used to re-sort

The screen spans four donors, and cross-donor reproducibility is informative about which high-ranking regulators are robust. We carry donor-robustness as an annotation column on the ranking (`hub_ranking_bayes.csv`), not as a re-sorting criterion. This is deliberate: re-ranking on donor concordance would silently hide the regulators that rank high but fail reproducibility, which is exactly the population a reader needs to see. As a column, it lets a reader see "ranked highly, but fails cross-donor concordance": SMG1, for example, stays at rank 24 in the primary ranking flagged `donor_robust = False`, rather than being silently demoted. That this matters is shown by the alternative — a reproducibility-aware re-ranking would push SMG1 from rank 24 to 288 (Figure S2) — so re-sorting on reproducibility would bury exactly the high-ranked-but-unreproducible regulators a reader most needs to see. We therefore keep the primary ranking (`hub_ranking_bayes.csv`) intact and carry reproducibility as the flag.

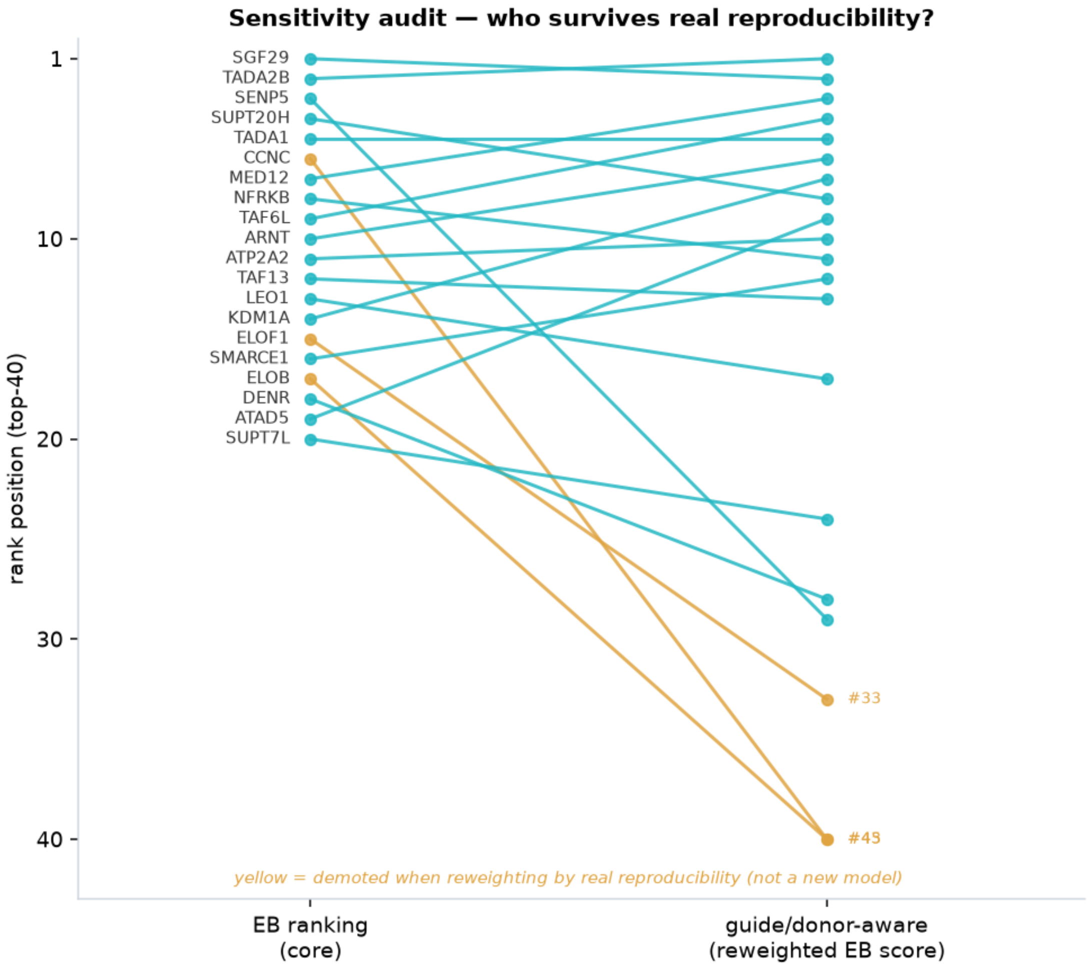


**Figure S2. Why donor-robustness is carried as a flag, not used to re-sort the ranking.** A reproducibility-aware re-ranking (right) demotes several top-40 regulators that fail cross-donor concordance (yellow) relative to the primary breadth-based ranking (left). The same reweighting pushes SMG1 from rank 24 to 288 — off the top-40 window shown here — which is exactly the high-ranked-but-unreproducible case a reader needs to see. We therefore keep the primary ranking (`hub_ranking_bayes.csv`) intact and flag donor-robustness (`donor_robust = False`) rather than re-sorting on it.

## The ranking is robust to cis-off-target exclusion

CRISPRi can knock down not only the targeted gene but its chromosomal neighbors, so an apparent regulator could be acting through a cis neighbor rather than the annotated target. The screen's `offtarget_flag` captures this, and we confirmed it is overwhelmingly a cis signal: it agrees with an independent neighboring-gene-knockdown call at 99.4%, of which the flagged cases are 92% cis. Excluding flagged regulators is therefore a legitimate specificity control. It does reshape parts of the ranking — the SAGA chromatin complex is the clearest case, where two subunits (TADA2B, TAF6L) prove cis-inflated and collapse under exclusion while the cis-clean subunit SUPT20H rises (rank 4 → 3). The complex's identity survives on its cis-clean members, but the leading-subunit list is corrected; we carry this correction forward into the program naming of Section 5, where SUPT20H alone anchors the SAGA program.

Taken together, these audits establish the ranking as an audited substrate: metric-robust, power-controlled, externally corroborated, transparent about donor reproducibility, and cleaned of cis-off-target artifacts. The operator built on these effect estimates in the following sections therefore inherits a controlled foundation rather than an assumed one.

## 3. A shared regulatory operator with a low-dimensional predictive subspace

Having established that the regulator ranking is trustworthy (Section 2), we now treat the screen not as a ranked list but as a single object: the regulator × gene × condition tensor of perturbation effects, which we call the regulatory operator. The question this section answers is whether that object has *shared* structure — structure that lets the measured perturbations inform the unmeasured response of others — or whether each regulator's effect profile is independent, in which case the operator is merely a stack of unrelated rows and the ranked-list view loses nothing.

### Construction

We assemble the operator as a regulator × gene × condition tensor in a pooled z-score space, where each entry is a knockdown effect standardized at the layer level rather than per condition. This choice is load-bearing: raw log-fold-change magnitude is confounded with statistical power (regulators assayed in more cells show larger apparent effects; Section 2), and a tensor built in raw space would encode that confound as structure. In the pooled z-score space the confound is removed — the rank correlation between per-regulator slab norm and cell count is $\rho = -0.006$ on the panel genes the decomposition uses, and a power gate on the singular subspace holds at a maximum $|\rho|$ of 0.049. We build on the set of regulators above a per-condition cell-count floor, which yields 3,106 regulators — a fourfold expansion over the 800-regulator pilot on which the approach was first validated. The confound guard is re-asserted after the expanded fetch and passes, so the fourfold-larger operator is estimated in the same clean representation as the pilot.

### The operator predicts the unobserved response of out-of-panel regulators

A first test of shared structure is whether observations in two conditions improve completion of a third. We hold out an entire condition fiber — the late-stimulation (Stim48hr) response — for regulators outside the original variance-selected panel, and ask whether a low-rank fit trained on the other regulators, reading only the resting and early-stimulation entries of the held-out regulators, recovers their held-out late-stimulation response. The baseline is persistence (late = early); its negative $R^2$ ($-0.305$ on the full held-out set) shows late stimulation genuinely diverges from early, so beating it means the fit is using cross-regulator structure to anticipate that divergence. This is evidence that the operator's structure is shared and usable — not the paper's central result, which is the modular organization recovered in Section 4.

On held-out out-of-panel regulators (~621 per holdout), the low-rank model beats persistence **at**

**every rank**, with an aggregate margin of $\Delta$ $R^2$ = **+0.382 ± 0.011** at rank 7 — mean ± SD over 20 independent random holdouts (source `operator_completion_multiseed_3106.csv`), confirming the margin is a stable property of the operator, not a single-seed artifact (Figure 2). Beating the baseline at every rank, rather than at a single tuned rank, is what makes the result a property of the operator rather than of a lucky hyperparameter.

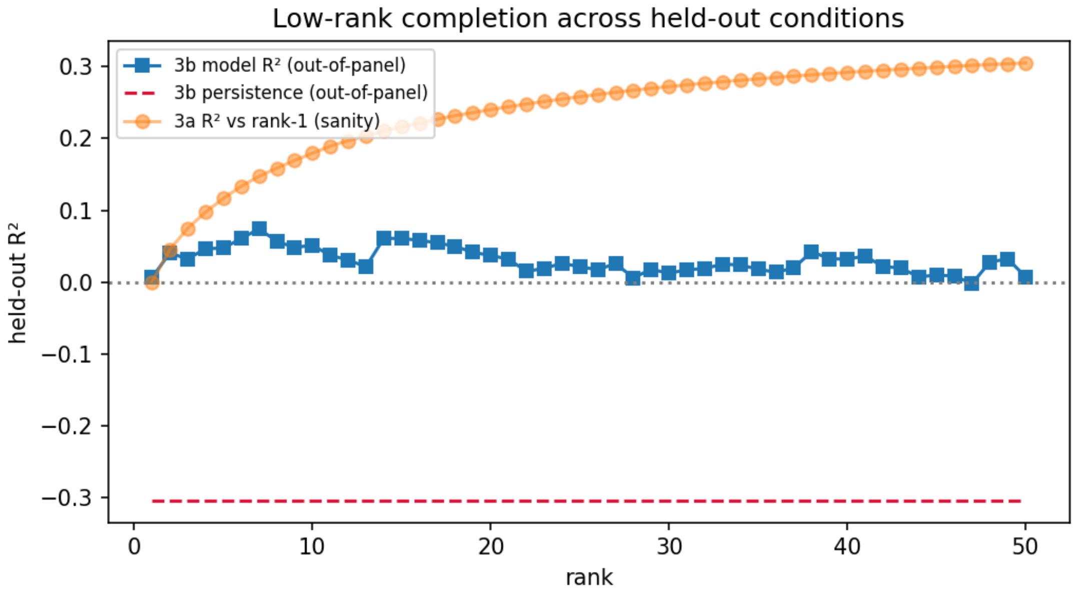


**Figure 2. The operator's predictive subspace is low-dimensional and beats persistence at every rank.** Out-of-panel completion $R^2$ for the low-rank fit versus the persistence baseline across ranks, on 621 held-out out-of-panel regulators of the 3,106-regulator operator. The low-rank model beats persistence at every rank; the aggregate margin peaks at rank 7 ($\Delta$ $R^2$ = +0.379) and declines thereafter — a genuine turnover that scopes the predictive subspace to ~7 dimensions.

The margin must be read carefully, and stratifying by knockdown strength keeps it honest (Table 1). It is *largest* for weak regulators (+0.501) and *smallest* for strong ones (+0.205 on the subset matched to the original panel's strength), each stable across the 20 holdouts — an ordering that is mechanistic, not mysterious: persistence is a fair baseline for strongly knocked-down regulators and collapses for weakly knocked-down ones, so the aggregate margin concentrates where the baseline is weakest. The conservative reading is the one we adopt: even for the strongest, panel-matched regulators the model still beats persistence (+0.205 at rank 7, ≥ +0.151 at every rank), so the advantage is not an artifact of weak-regulator baseline collapse.

**Table 1. The out-of-panel predictive advantage survives strength stratification.** Held-out late-stimulation completion at rank 7 (the peak rank), by knockdown-strength stratum, read from `operator_completion_stratified_3106.csv` (a single representative holdout, seed 0; margins are stable across 20 random holdouts, mean ± SD in `operator_completion_multiseed_3106.csv`). The margin is largest for weak regulators, where the persistence baseline collapses, but stays clearly positive even for the strongest, panel-matched regulators — for which persistence is a fair baseline — so the advantage is not an artifact of weak-regulator baseline collapse.

| Stratum | $n$ | Model $R^2$ | Persistence $R^2$ | Margin $\Delta$ $R^2$ |
|---|---|---|---|---|
| All out-of-panel | 621 | 0.074 | −0.305 | **+0.379** |
| Strong (median split) | 311 | 0.069 | −0.208 | +0.277 |
| Weak (median split) | 310 | 0.080 | −0.421 | +0.501 |
| Strong, panel-matched | 203 | 0.058 | −0.146 | +0.205 |

Two boundaries on this claim are worth stating explicitly, because both are places a reader could otherwise over-read the result. First, the held-out unit is a condition *fiber*, not a whole regulator: each held-out regulator is observed in the resting and early-stimulation conditions, and only its late-stimulation response is predicted. A purely low-rank model cannot predict a regulator with no observed entries at all, and we make no such claim — the result is extrapolation across a regulator's conditions, not imputation of never-measured regulators. Second, the absolute predictive precision is modest: the model's held-out $R^2$ is in the range 0.06–0.08 across all strata, well below the 0.154 the pilot reached on its smaller 160-regulator holdout. The larger, more heterogeneous 3,106-regulator population makes the shared low-rank structure a modestly worse per-regulator predictor in absolute terms, even at matched strength. The beats-persistence *advantage* is robust and reproducible across strata and ranks; the absolute precision is modest and does not match the pilot. We report both halves, because reporting only the first would be an inflated headline.

## The predictive subspace is low-dimensional

The rank at which prediction is best is itself informative. The out-of-panel completion $R^2$ peaks at rank 7 and declines thereafter through rank 50 (Figure 2) — a genuine turnover, not a curve still climbing at the edge of the swept range. This licenses a low-dimensionality statement, but a carefully scoped one: it is the **predictive subspace** that is low-dimensional (~7 components), not the operator as a whole. The operator's own effective rank, measured on its singular spectrum, is roughly an order of magnitude higher (Figure S1); the ~7-dimensional structure is the part of it that generalizes to out-of-panel regulators. The distinction matters — a reader who took "low-rank operator" at face value would expect a spectrum that collapses after seven components, and the spectrum does not. What is low-rank is the shared, transferable structure, and it is a property of this 3,106-regulator holdout configuration rather than an intrinsic constant of T-cell regulation.

## The representation stays pooled at scale

A fourfold expansion of the regulator axis could in principle break the pooling that makes the operator one object rather than many: if the added regulators occupied their own disjoint subspace, the "shared" structure would be an average over unrelated blocks. A CANDECOMP/PARAFAC decomposition of the expanded operator rules this out. Two factors are gated by condition with bootstrap confidence intervals excluding a flat profile (peaking in resting and in early-stimulated cells respectively; Section 5), and two further factors are cleanly constitutive — a cleaner decomposition than the pilot, which recovered no clean constitutive factor. That the decomposition finds coherent, condition-structured, partly-shared factors at 3,106 regulators is positive evidence that the representation remains genuinely pooled at the expanded scale, not a mosaic of independent per-regulator effects.

Taken together, these results establish the operator as a shared structure that supports condition completion: a low-dimensional structure that anticipates the unobserved late-stimulation response of regulators outside the panel, robustly beats a fair baseline, and remains one pooled object at four times the scale at which it was first validated — with an absolute precision we report honestly as modest.

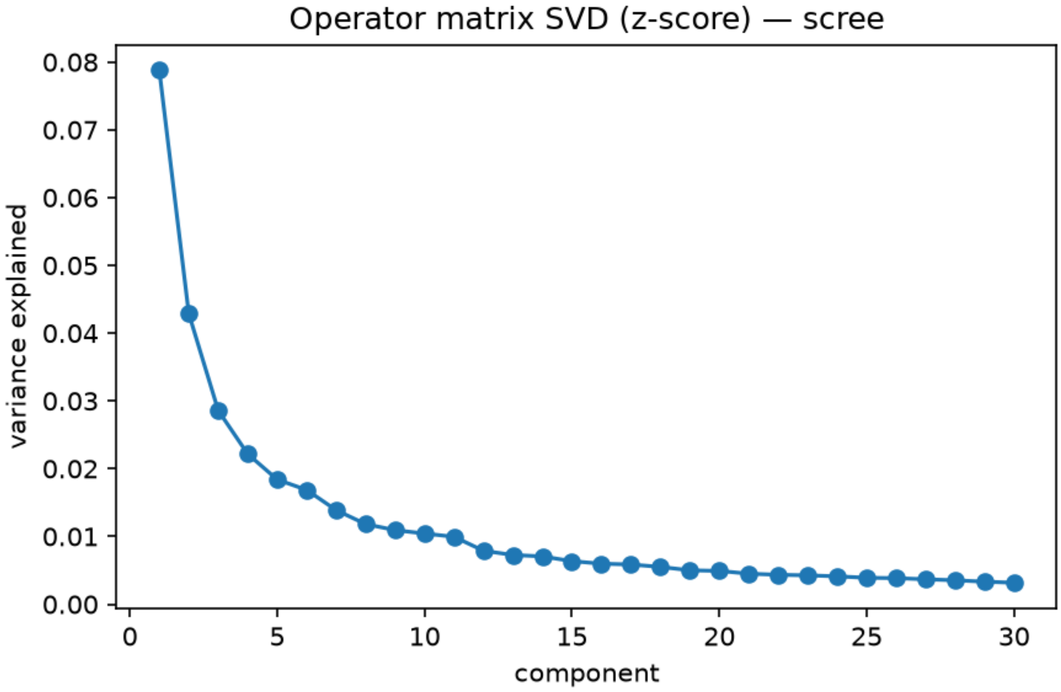


**Figure S1. The operator's own effective rank is far higher than its predictive subspace.** Singular-value scree of the 3,106-regulator operator. The spectrum does not collapse after seven components (effective rank ≈ 86); the ~7-dimensional structure identified in Figure 2 is the transferable part that generalizes to out-of-panel regulators, not the operator's full rank.

## 4. Denoised operator structure reveals regulatory modules

The preceding sections establish which effects are trustworthy: regulators identified above a reproducibility- and context-audited bar, not raw hubs or nominal significance. This section asks what those trustworthy effects build when taken together. The operator induces a regulator–regulator similarity, and that similarity is not structureless — it carries discrete, co-regulated modules. This is where the analysis changes scale, from individual regulators to the organization of the system, and it is the strongest use of the shared structure in this work: we recover the modules blind to any annotation, and only then ask what they are (Figure 3).

### 4.1 Spectral denoising isolates low-dimensional regulatory structure

Each regulator is represented by its 2,000-gene z-score fingerprint with the three conditions concatenated (3,106 × 6,000, aspect ratio $q = N/T = 0.52$), and the regulator–regulator correlation of these fingerprints is the object we cluster. Raw, it is dominated by two nuisance structures that random-matrix theory isolates cleanly. A single global co-variation mode (leading eigenvalue $\lambda_0 = 167$, an order of magnitude above the rest) is the shared perturbation-response axis loading every regulator; it is deflated. The Marchenko–Pastur noise bulk (fitted edge $\lambda_+ = 1.49$ after deflation) is discarded. What remains — the correlation reconstructed from the signal eigenvalues between the bulk edge and the global mode — is the denoised operator.

The count of genuine signal directions must be set honestly, because the closed-form MP edge is a *lower bound*: the 6,000 feature columns are not independent (the same gene appears in all three conditions), so the effective sample size is below 6,000 and noise eigenvalues leak upward. An empirical null fixes this — the 95th percentile of the top eigenvalue across 100 column-

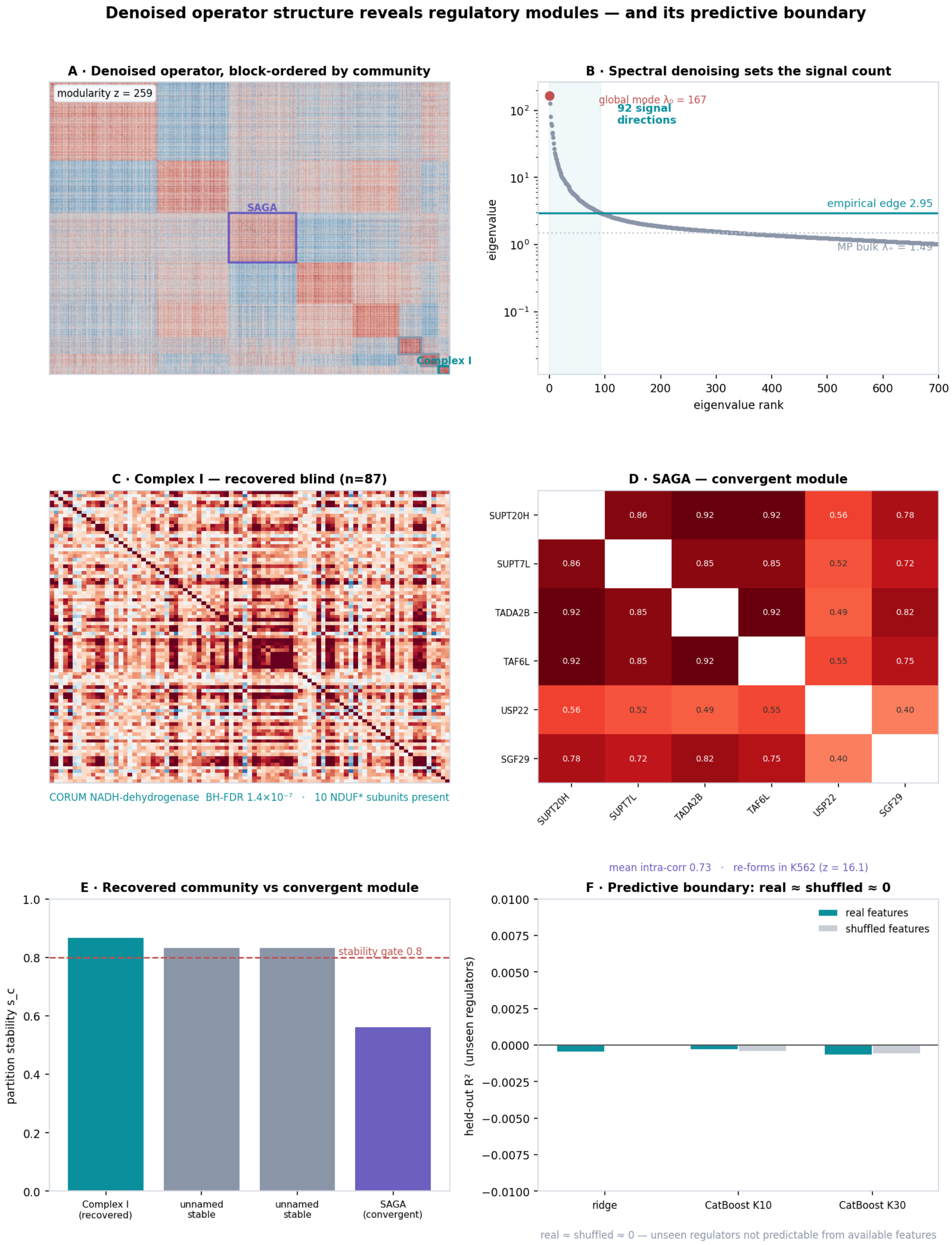


**Figure 3. Denoised operator structure reveals regulatory modules — and the boundary on prediction.** (A) The denoised regulator–regulator operator, block-ordered by community, with the SAGA and Complex-I communities outlined (consensus modularity z = 259 against a label-permutation null). (B) Eigenspectrum: after deflating the global co-variation mode ($\lambda_0 = 167$) and discarding the Marchenko–Pastur bulk ($\lambda_+ = 1.49$), an empirical column-permutation edge (2.95) leaves 92 signal directions rather than the 336 the closed form admits. (C) The stability-gated community matching mitochondrial Complex I, recovered blind to annotation (CORUM NADH-dehydrogenase, BH-FDR $1.4 \times 10^{-7}$; 10 NDUF* subunits present). (D) The SAGA subunits' operator correlations — a convergent module whose subunits also re-form a cluster in K562 (z = 16.1). (E) Partition stability against the 0.8 gate distinguishes a recovered community (Complex I) from the convergent SAGA module (0.56, below the gate). (F) Leave-regulator-out prediction of unseen regulators: real and shuffled features perform identically at $R^2 \approx 0$ (ridge and CatBoost), so the negative rests on the shuffled control — the predictive boundary.

permuted correlations gives an edge of $\lambda_+ = 2.95$, leaving **92 signal eigenvalues, not the 336 the closed form would admit**. We use 92 throughout. The point of this subsection is that the clustering below runs on genuine low-dimensional structure, not on noise the denoising failed to remove.

### 4.2 The denoised operator contains highly non-random community structure

Consensus community detection (Leiden with an RBConfiguration objective, swept over resolution and 50 random seeds, 500 partitions consensus-aggregated) yields eight communities, three of which pass a partition-stability gate of $s\_c \geq 0.8$. The structure is overwhelmingly non-random: the consensus modularity Q = 0.506 sits **z = 259** above a label-permutation null. This is a property of the denoised operator, established before any community is named — the community structure is strongly supported relative to the matched null, and the question of what the modules correspond to is therefore worth asking.

### 4.3 Blind community detection recovers mitochondrial Complex I

The cleanest result in this work is what the clustering produces with no annotation in the loop. Of the three stability-gated communities, one (n = 87, 45% donor-reproducible) matches **Mitochondrial Respiratory Chain Complex I** (NADH dehydrogenase) under CORUM enrichment at BH-FDR $1.4 \times 10^{-7}$ — 8 of the 13 CORUM holoenzyme members, with 10 NDUF-family genes present in the community overall, corroborated across seven Complex-I assembly and intermediate entries. **An annotation-blind partition recovered a known molecular complex.** The community was defined purely by co-movement of perturbation signatures across the transcriptome and only afterward matched to a database; nothing about its construction knew the subunits belonged together. This single result validates the denoising, the fingerprint representation, and the clustering at once — a stable, unsupervised community corresponding exactly to a curated complex is not something noise or a flexible interpretation can produce.

That a mitochondrial electron-transport module surfaces coherently under $CD4^+$ perturbation is expected biology, not a housekeeping artifact: ETC respiration is induced and required for T-cell activation, proliferation, and memory formation (Tarasenko et al., *Nat. Commun.* 2025), so knocking down its subunits produces one coordinated, activation-relevant transcriptional response — which is why the subunits co-cluster into a stable, donor-robust community. The other two stability-gated communities (n = 177 and 132) match no annotated CORUM complex at FDR $< 0.05$: real, reproducible regulator modules the atlas has not otherwise surfaced, reported as such rather than named by force.

### 4.4 Convergent evidence identifies a SAGA-centered regulatory module

The second module is a richer inference resting on a different, and honestly weaker, evidentiary mode. Six SAGA chromatin-coactivator subunits (SUPT20H, SUPT7L, TADA2B, TAF6L, USP22, SGF29) fall in one community, with only KAT2B elsewhere — but this community's partition stability is 0.56, *below* the 0.8 gate. We therefore do not claim that stable community detection recovered SAGA. Instead the SAGA-centered module rests on three independent lines converging on the same grouping. CORUM enrichment names it: GCN5-linked SAGA at BH-FDR $3 \times 10^{-4}$ (8/8), STAGA at $10^{-3}$, core SAGA and Mediator below 0.05. The CP factors carrying SAGA loadings (factors 2, 3, 5) map to it at 100% concentration. And rebuilding the regulator–regulator correlation *in K562* from Replogle's independent screen, the same subunits re-form a coherent cluster — mean intra-correlation 0.244 against $0.010 \pm 0.014$ for random same-size sets ($z = 16.1$, $p < 10^{-4}$), with five of the six shared subunits scoring universal in the cross-cell-type concordance of Section 6.

The distinction from Complex I is deliberate and we keep it visible throughout: Complex I is a **recovered community** — isolated directly by stable, annotation-blind clustering — while SAGA is a **convergent regulatory module**, supported by agreement across CORUM identity, CP-factor concentration, and cross-cell-type cohesion, but not by a sufficiently stable Leiden partition. Stating the 0.56 openly is what licenses the convergent-evidence claim; conflating the two under one "community" label would blur a real methodological difference.

### 4.5 Module structure nominates disease-linked regulators

Known biology validated the structure; the module organization can, in turn, generate nominations. Because the communities are groups of T-expressed regulators, we ask whether any carries more autoimmune genetic-risk signal than chance — as a nomination, not a causal claim, against a null built to avoid the obvious trap. Immune genes are long, SNP-dense, and clustered near the MHC, so a genome-wide null is trivially "enriched"; we instead resample 10,000 length-decile-matched sets from the 3,106 panel regulators themselves, reported with and without the chr6 MHC block (Open Targets genetic_association, five autoimmune diseases). The controls validate the null exactly: the recovered Complex-I community is not enriched (fold 0.45, $p = 0.98$) — a leaky null would falsely light it up — while the SAGA-centered module is enriched (fold 1.35, $p = 0.0025$, surviving MHC exclusion) broadly across four diseases (RA/SLE/MS/T1D loads 15.1/8.4/8.3/9.7), including canonical T-cell risk regulators GATA3, CBLB, UBASH3A, IL12RB2, and RASGRP3. The two stable unnamed communities carry no signal above the matched null (folds 0.97 and 1.31; $p = 0.55$ and 0.10). The coactivator module that organizes T-cell activation touches autoimmune risk genes; the metabolic module does not — a nomination a follow-up can act on, illustrating what the validated structure enables rather than promising a clinical result.

## 5. The operator decomposes into gene programs gated by activation state

Section 3 established that the operator has shared, low-dimensional predictive structure. This section asks what that structure *is*: when the operator is decomposed into factors, do those factors correspond to coherent gene programs, and are those programs organized by the biological variable the screen was designed around — the cell's activation state? We decompose the operator with a CANDECOMP/PARAFAC (CP) factorization, which represents it as a sum of rank-one components, each carrying a regulator loading, a gene loading, and a condition-modulation profile. The condition profile is what makes CP the right tool here: it reads off, for each program, how its transcriptional magnitude varies across resting, early-stimulated (Stim8hr), and late-stimulated (Stim48hr) cells.

### Condition gating is the proven result

Two factors are gated by activation state with bootstrap confidence intervals that exclude a flat profile (Figure 4; source `operator_cp_factors_3106.csv`). One peaks in resting cells (factor 1: condition weights 0.91 / 0.39 / 0.15 across Rest / Stim8hr / Stim48hr) and one peaks in early stimulation (factor 6: 0.08 / 0.98 / 0.17). Two further factors are cleanly constitutive — flat across conditions; the remaining two (factors 2 and 5) form a degenerate pair (cofactor cosine

0.97) and are excluded from interpretation. That both a resting-peaked and a stimulation-peaked program emerge, each with a confidence interval excluding flatness, is the load-bearing claim of this section: the operator's shared structure is not a single undifferentiated block but resolves into programs whose activity is *modulated by* activation state, in both directions. This is a proven, bootstrap-supported statement about condition identity, and it is what clause (ii) of the thesis claims — nothing more.

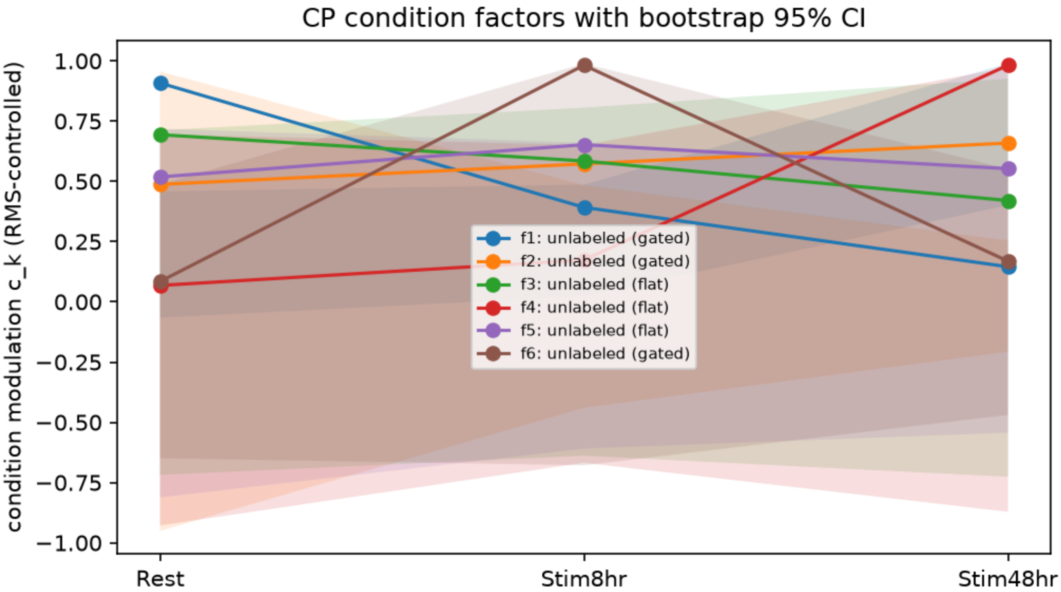


**Figure 4. The operator decomposes into gene programs gated by activation state.** CANDECOMP/PARAFAC condition-factor weights across Rest / Stim8hr / Stim48hr. Two factors are gated with bootstrap confidence intervals excluding a flat profile — one peaking in resting cells (factor 1) and one in early stimulation (factor 6) — while two further factors are constitutive, evidencing bidirectional gating rather than a single activation program. The remaining two lines (factors 2 and 5) are a degenerate pair (cofactor cosine 0.97) excluded from interpretation, so the interpreted set is four factors — two gated, two constitutive.

We are deliberate about the word "modulated" rather than "activated": the two gated programs move in opposite directions with stimulation. Reading these as "the activation programs" would miss that one of them is a resting-cell program that stimulation turns *down*. The decomposition recovers bidirectional gating, and the prose should not flatten it into a one-directional activation story.

### Naming the programs — non-circular, and honestly thin where it is thin

Assigning a biological name to a factor is a separate, weaker step than proving it is gated, and we hold it to two independent filters. First, non-circularity: a regulator is assigned to a factor by its loading in the CP fit, which is computed with no access to any biological annotation — so recognizing that a factor's top regulators form a known complex is a genuine test, not a tautology. Second, cis-cleanliness: a regulator can anchor a name only if it survives the cis-off-target gate of Section 2, because a regulator whose apparent effect came from knocking down a neighboring gene is not evidence for anything.

Under both filters, **factor 6 reads as an early-activation program coupled to SAGA chromatin machinery.** Its top regulators include three SAGA subunits — TADA2B, TAF6L, and SUPT20H — but here the cis gate does real work: TADA2B and TAF6L are cis-inflated, collapsing from ranks 2 and 9 to 93 and 130 under a hard cis-off-target exclusion, so the SAGA anchor rests on the one cis-clean subunit present, **SUPT20H**. This makes factor 6's SAGA signal thinner than its raw subunit list suggests — cis-clean, it is essentially one subunit — and we state it that way rather than citing the full list. Its gene loadings are enriched for early T-cell-activation effector transcripts (leukocyte actin- and TCR-signaling effectors — CORO1A, ARPC1B, GPSM3, LIMD2, MYL12A; empirical "TCR signaling|up" markers, hypergeometric p = 1.9e-5), consistent with the Stim8hr peak. The curated TCR-proximal kinase set (ZAP70/LCK/LAT) shows no overlap with factor 6, as expected — these are downstream readout genes, not proximal signaling components.

**Factor 1 reads as a resting-gated program spanning proteostasis/RNA-processing on the gene side and a multi-module regulator side.** Its downstream genes co-load chaperone and RNA-processing transcripts (TCP1, HSPD1, HSPE1, HNRNP family); because the repository carries no curated proteostasis gene set, this is an identity-and-cross-decomposition call, not a hypergeometric p-value, and we label it as such. On the regulator side, factor 1 collects two recognizable modules: the AHR–ARNT xenobiotic-sensing heterodimer, and the immune Rac→WAVE→actin cytoskeletal axis (NCKAP1L/Hem1, WASF2, DOCK2, ARHGAP30). Whether these two modules are one program or two that the rank-6 truncation fuses is a question we can pose but not settle: they co-load at the stable rank and separate only at ranks 8–10, which sit below the decomposition's stability floor, so the split is suggestive, not decisive — we cannot distinguish two genuine programs emerging from ordinary factor rotation in an unstable regime. We report the co-loading and stop there.

The honest summary of this section is a two-tier one: condition gating is proven with bootstrap CIs; the program *names* are a weaker, filtered overlay, strongest for factor 6's SUPT20H-anchored SAGA coupling and explicitly tentative for factor 1's internal structure. The next section tests one of these names against an entirely independent axis of evidence.

## 6. Generalization: specificity across cell types, and the predictive boundary

The programs named in Section 5 are a within-dataset result: they come from decomposing the CD4$^+$ T-cell operator alone. This section tests one of the strongest naming claims — that the immune Rac→WAVE→actin cytoskeletal module is T-cell biology rather than generic cellular machinery — against an entirely independent axis: how the same regulators behave in a different cell type. If a regulator's perturbation signature is conserved between CD4$^+$ T cells and an unrelated lineage, its regulation is likely universal cellular housekeeping; if the signature diverges, the regulation is cell-type-specific. The comparator is the genome-wide Perturb-seq screen of Replogle et al. (2022) in K562, a chronic-myelogenous-leukemia line with no T-cell identity.

### Construction and controls

We use Replogle's pre-aggregated, Z-normalized bulk product — a perturbation × gene signature matrix, the correct unit for a per-regulator concordance, rather than the 8.8 GB cell-level file. Regulators are collapsed to gene level and non-targeting controls dropped (602 removed), genes are matched by Ensembl identifier, and 6,407 regulators are shared between the two screens. For each shared regulator we compute a per-regulator Pearson concordance between its Z-scored CD4$^+$ and K562 effect profiles over co-measured genes, and convert it to a Z-score against a per-regulator null.

Three controls guard the split. A label-permutation null — permuting the cross-dataset pairing so each CD4$^+$ regulator is compared against a random K562 regulator — is centered at zero with a 95th percentile of 0.038, so the observed concordances are not a baseline artifact. A coverage check finds essentially no dependence of concordance on the number of co-measured genes (Spearman −0.010; the data are near-complete), so missingness drives nothing. The decisive control is for statistical power: concordance correlates with knockdown strength (Spearman 0.200 against the minimum per-side knockdown rank; source `k562_concordance.csv`, `min_kd_rank` column), meaning a regulator can look "T-specific" simply because it was weakly perturbed in one screen. We therefore do not report the raw split; we lead with the well-powered subset, where both screens knocked the regulator down strongly.

### The result, and the inversion

Among well-powered regulators, 508 are universal and 227 are T-cell-specific; restricting further to the donor-reproducible core gives 201 universal and 49 T-specific. Both are conservative counts, and deliberately so — for immune regulators that are lowly expressed in K562, a weak K562 knockdown *is itself* a form of cell-type specificity, so the power filter discards some genuine T-specific signal along with the noise. The T-specific count is a floor, not an estimate.

The biological pattern inverts the naive expectation. One might guess that the activation-coupled program (factor 6) would be the T-cell-specific one and the constitutive machinery universal; the data give the opposite, and it is more coherent for being so. **SAGA chromatin machinery is universal**: SUPT20H — the cis-clean SAGA anchor from Section 5 — is concordant between the two cell types (Z = +0.186, well-powered and donor-reproducible). Chromatin/transcription machinery does the same job in any cell, so its conservation is expected. **The immune Rac→WAVE→actin module is T-cell-specific**: its clean anchor is DOCK2, a lymphocyte Rac guanine-nucleotide-exchange factor, which is well-powered and donor-reproducible and shows no cross-cell-type concordance. This is the same module Section 5's decomposition assigned to factor 1 — so two independent analyses, the within-dataset CP decomposition and the cross-cell-type comparison, converge on the identity of the immune cytoskeletal module as T-cell-specific.

We anchor each arm on a single clean regulator, symmetrically, because that is what the evidence supports. On the universal side, SUPT20H alone survives every filter — TADA2B and TAF6L also score universal but are cis-inflated (Section 5) and not well-powered. On the T-specific side, DOCK2 alone is well-powered; the module's other members — NCKAP1L (Hem1), AHR, and ARNT — are donor-reproducible and classified T-specific but fall below the well-powered threshold, so their T-specific label rests on the conservative-floor argument rather than on independent power. The strong form of the claim is therefore carried by one clean anchor per arm, SUPT20H and DOCK2, with the surrounding modules consistent but not independently powered — the same evidentiary discipline applied on both sides.

The value of this section is that it converts a within-dataset naming call into a cross-dataset, externally validated distinction: the operator does not merely factor into programs, it separates regulation that is generic to cells from regulation that is specific to the T-cell lineage, and it does so in agreement with an independent screen the decomposition never saw.

### Specificity is sustained across a third cell type (RPE1)

The universal-versus-specific distinction drawn above rests, so far, on two cell types — primary CD4$^+$ T cells and leukemic K562 — both of them extremes. A single pairwise comparison cannot separate a genuine lineage-specificity axis from an idiosyncrasy of one contrast. We add a third, genuinely distinct context: RPE1, a non-cancerous, hTERT-immortalized, near-euploid, p53-competent epithelial line, using Replogle's essential-scale screen and the identical concordance pipeline. Because RPE1 is essential-scale rather than genome-wide, we gate on regulator overlap before proceeding — 962 regulators are shared, past the pre-registered threshold, with clean power and coverage controls.

The distinction is sustained, asymmetrically and honestly. SAGA reappears universal: of the three SAGA subunits present in the essential-scale set, two clear the null (TADA2B and TAF6L, $z = +0.163$ and $+0.150$ against $q95 = 0.121$) and the third sits at threshold (SUPT20H, $z = +0.117$) — the K562 pattern, now in a third lineage. The T-cell-specific side is carried at the class level: of the 142 regulators classified T-specific in the CD4↔K562 comparison and also present in RPE1, none become universal (mean concordance $z = -0.013$). We flag the honest limit rather than paper over it: DOCK2, the clean T-specific anchor, is not in RPE1's essential-scale screen at all — it is a hematopoietic-specific exchange factor, not a core-essential gene — so its arm is carried by the class-level result, not a direct measurement; its absence is consistent with the essential-scale design and prevents direct validation of the anchor in RPE1. Across three diverse backgrounds, SAGA regulation reads as universal and the immune cytoskeletal module as T-cell-specific — a distinction built from data in three lineages rather than asserted from one.

### Operator structure does not imply predictability for unseen regulators

That the operator has compressible structure does not mean that structure can be predicted, for a new regulator, from external covariates. We tested this directly. Holding out whole regulators — all their conditions at once — we asked whether their trans-response could be predicted from side-features alone: transcription-factor and DNA-binding-domain annotation, protein–protein network connectivity, and functional-category composition, covering 98–99% of the panel.

Real and permuted regulator features yielded indistinguishable performance, indicating that the available annotations contain no detectable signal for leave-regulator-out prediction. A linear model (multi-output ridge, $R^2$ across five folds between $-0.0015$ and $+0.0002$) and a non-linear gradient-boosted learner on the response's leading components ($R^2$ between $-0.0035$ and $+0.0010$, at 10 and 30 components alike) agree: out-of-sample $R^2$ is statistically indistinguishable from zero, and — decisively — from the same model trained on features permuted against the responses (shuffled control $\approx 0$ throughout). When real and scrambled inputs perform identically, there is no learnable map to recover. Per a pre-registered criterion, a model that fails to beat its shuffled control ends the inquiry rather than motivating a larger one, so we did not pursue further model classes.

This is a conceptual boundary, not a tuning failure, and it is consistent with the field's finding that current perturbation foundation models do not beat a train-mean predictor on unseen perturbations (Ahlmann-Eltze et al. 2025). Low-dimensional organization is not equivalent to out-of-sample predictability: the operator's structure is rich enough to group known regulators into reproducible modules and to extrapolate across conditions for regulators it has seen, yet the map from a regulator's identity to its trans-effects is not recoverable, on this data, from the covariates available. We state the boundary rather than obscure it.

## 7. Discussion

### What the operator view buys

The main result of this work is that a quality-controlled perturbational operator contains recoverable modular organization beyond noise and hub structure. After spectral denoising isolates roughly ninety genuine signal directions from an atlas of thousands of regulators, the surviving regulator–regulator structure is strongly non-random and resolves into discrete co-regulated modules — organization that is invisible to a ranked hit list and that emerges only when the screen is treated as a single object.

Complex I provides the cleanest validation, precisely because it was recovered without annotations. A stability-gated community, defined purely by co-movement of perturbation signatures, matches a curated molecular complex at FDR $1.4 \times 10^{-7}$; the clustering never saw the subunit assignments. When an unsupervised partition reconstructs a known complex, the denoising, the fingerprint representation, and the clustering are validated simultaneously — there is no interpretive degree of freedom to absorb the result.

SAGA illustrates a different evidentiary mode, and the contrast is worth stating plainly because it reflects methodological maturity rather than a weakness to hide. The SAGA-centered module is not a stably recovered community — its Leiden partition stability sits below our gate — but three independent sources converge on the same grouping: database identity, factor concentration, and cohesion in a second cell type. We distinguish these two classes of discovery — a *recovered community* versus a *convergent regulatory module* — rather than merge them under one label, because the difference is real and a careful reader would otherwise find it themselves.

The same object is what makes the rest of the analysis possible: it predicts the unobserved late-stimulation response of out-of-panel regulators from shared low-rank structure (Section 3), reads condition-gating off directly as bootstrap-tested factors (Section 5), and turns each regulator's effect profile into the unit of a cross-cell-type comparison sustained across three lineages (Section 6). But recoverable structure is not predictive reach. Holding out whole regulators, real and scrambled side-features perform identically, so the map from a regulator's identity to its trans-effects is not recoverable from available annotations. The operator organizes what it has seen; it does not, on this data, generalize to what it has not — a boundary we make explicit rather than eliding.

### Honest limits

We gather the caveats here rather than leave them distributed, because together they define the scope of the claims. First, low-rank is a property of the *predictive subspace*, not the operator: the completion optimum at ~7 components describes the transferable structure, while the operator's own effective rank is roughly an order of magnitude higher. A reader should not take "low-rank operator" as a description of the object — only of the part of it that generalizes. Second, the program names rest on single clean anchors. After the cis-off-target and power filters, the SAGA program is anchored on SUPT20H alone and the T-cell-specific cytoskeletal module on DOCK2 alone; the surrounding subunits and modules are consistent but not independently powered, and we have been explicit about which regulators carry the claim and which ride along on a conservative-floor argument. Third, absolute predictive precision is modest and softens at scale: the out-of-panel model $R^2$ is 0.06–0.08 across strata, below the 0.154 the smaller pilot reached, so the robust finding is the *advantage* over baseline, not a high absolute accuracy.

Beyond these bounded caveats there is one genuine conceptual gap. We inherit the effect estimates from the screen's harmonized differential-expression pipeline rather than jointly modeling the raw counts with guide-assignment uncertainty — the territory of calibrated single-cell-CRISPR testing and guided factor models. A fully generative treatment that propagated assignment uncertainty into the operator would be more principled, and we defer it deliberately: it requires the cell-level data (1.8 TB) and the compute that entails, against a design goal of a laptop-reproducible pipeline. Naming this gap is part of the honest accounting, not a claim that our shortcut is free.

### Future directions

Three extensions follow naturally. The donor axis can be examined as a subspace-stability question — whether the operator's leading structure is preserved across donors, measurable as principal angles between per-donor operators — turning donor reproducibility from a per-regulator annotation into a statement about the shared structure itself. The operator's off-diagonal structure invites a network-deconvolution treatment, which should be benchmarked against the recent single-cell perturbation network-inference benchmark of Chevalley et al.[9] rather than asserted, given that field's finding that inference methods often fail to beat simple baselines. And the low-power regulator tail, excluded here by the cell-count floor, could be recovered with a precision-weighted (lfcSE) sub-floor build if those specific regulators — rather than the well-powered structure — became the target of a follow-up.

The contribution is a way of reading genome-scale perturbation data as an auditable operator whose geometry contains recoverable modular and context-dependent biological organization, while placing a measured boundary on what that organization can predict — recovered with explicit statistical rigor from a 1.8 TB atlas, on a laptop.

## 8. Methods

### Dataset and access

We analyze the genome-scale CRISPRi Perturb-seq screen in primary human $CD4^+$ T cells (four donors; three activation states: resting, Stim8hr, Stim48hr). The full atlas is ~1.8 TB; all analyses in this paper run without loading the cell-level data. Instead we use the published supplementary differential-expression tables (~15 MB), a remote per-condition z-score slice fetched on demand (~0.1 GB, cached locally), and, for the cross-cell-type comparison, the Replogle et al. (2022) pre-aggregated K562 bulk product (~375 MB). The entire pipeline is therefore reproducible on a laptop; no step requires the cell-level `.h5ad`/`.h5mu` files.

### Effect estimates and the power confound

Per-regulator effect profiles are the calibrated differential-expression estimates from the screen's harmonized pipeline; we do not re-derive them. Because raw log-fold-change magnitude is confounded with statistical power — regulators assayed in more cells show systematically larger apparent effects (Spearman $\rho$ between per-regulator effect-vector norm and cell count $= -0.68$ in raw space) — all operator analyses are performed in a pooled z-score space, in which each entry is standardized at the layer level across the full regulator population rather than per condition.

## Operator tensor construction

The operator is assembled as a regulator × gene × condition tensor. Genes are the top-variance panel (2,000 genes by default). Regulators are selected by a per-condition cell-count floor (retaining regulators above the median), which yields 3,106 regulators for the escalated analysis and 800 for the validation pilot. After the z-score fetch, a confound guard re-measures the rank correlation between per-regulator slab norm and cell count on the panel genes and refuses to cache the tensor if $|\rho| \geq 0.15$ (fail-closed); the escalated tensor passes at $\rho = -0.006$, and a power gate on the leading singular subspace holds at max $|\rho| = 0.049$.

## CP decomposition

The tensor is decomposed with a masked CANDECOMP/PARAFAC factorization (`tensorly parafac` with a missingness mask, `n_iter_max = 400`, `n_init = 10`, `random_state = 0`). Before fitting, condition slabs are RMS scale-controlled so that a single high-variance condition cannot dominate the fit. Rank is selected by split-half stability: for each candidate rank we compute the matched-factor cosine between decompositions of two random halves (subsample 400), and take the largest rank whose stability exceeds 0.70. Each factor carries a regulator loading, a gene loading, and a three-entry condition-modulation profile; condition gating is assessed by bootstrap (100 resamples over conditions), and a factor is called gated only if the bootstrap confidence interval on its modulation profile excludes a flat profile. Degenerate factors — those with a maximum cross-factor cofactor cosine near 1 (≈0.97) — are excluded before interpretation. Factor loadings are varimax-rotated (`gamma = 1.0`, convergence on rotation change, `tol = 1e-6`).

## Out-of-panel completion

Predictive structure is assessed by low-rank matrix completion of a held-out condition fiber. We hold out the late-stimulation (Stim48hr) entries of regulators outside the original variance-selected panel (a seed-0 20% sample), fit a low-rank model to the remaining entries, and predict the held-out fiber. Standardization is computed on the training entries only (centering each gene on its train-mask values) to prevent leakage. Completion uses soft-impute with a truncated SVD (`n_iter = 100`, `tol = 1e-4`); the truncated solver is numerically identical to a full-SVD soft-impute to machine precision (a regression test asserts agreement to ~1e-13) and is roughly an order of magnitude faster. The baseline is persistence — predicting the late-stimulation response equals the early-stimulation one. We report the model—persistence margin per rank, and stratify the held-out regulators by knockdown strength, with the stratum thresholds documented alongside the results: a median split on effect strength, and a panel-matched stratum (regulators at least as strong as the median in-panel regulator).

## K562 cross-cell-type concordance

Cell-type specificity is assessed against the Replogle et al. (2022) genome-wide K562 Perturb-seq screen, using its pre-aggregated, Z-normalized perturbation × gene bulk product (`K562_gwps_normalized_bulk_01.h5ad`). Regulators are collapsed to gene level with an unweighted mean over guides/promoters, non-targeting controls are dropped (602 removed), and genes are matched by Ensembl identifier, yielding 6,407 shared regulators. For each shared regulator we compute a Pearson concordance between its Z-scored $CD4^+$ and K562 effect profiles over co-measured genes (no imputation of non-measured entries), converted to a per-regulator Z-score. Three controls are applied: a label-permutation null (permuting the cross-dataset pairing; 95th percentile 0.038); a coverage check (Spearman between concordance and number of co-measured genes, −0.010); and a power control (Spearman between concordance and the minimum per-side knockdown rank, 0.200 via the `min_kd_rank` column). Because concordance depends on power, we lead with the well-powered subset (both screens knocking the regulator down strongly) and report the donor-reproducible core as a conservative floor. Regulators are classified universal / T-cell-specific / intermediate on the sign and magnitude of their concordance Z-score.

## Reproducibility

The full operator pipeline runs end-to-end from a single build command (tensor → SVD → CP → completion → donor angles). Numerical kernels are isolated in a tested module (17 unit tests), including the truncated-SVD/full-SVD equivalence regression test. Random seeds are fixed (`random_state = 0`) throughout. All result tables cited in the text are written to `docs/tables/` with a `_3106` suffix for the escalated analysis, and the pilot-scale tables are retained alongside them so the pilot↔escalation comparison is itself reproducible from disk.

## Spectral denoising and community detection

Each regulator's 2,000-gene z-score fingerprint was concatenated across the three conditions (3,106 × 6,000; q = 0.52) and the regulator–regulator Pearson correlation formed. The leading global mode ($\lambda_0 = 167$) was deflated and the Marchenko–Pastur bulk (edge $\lambda_+ = 1.49$) discarded; the denoised operator is reconstructed from the intermediate signal eigenvalues. Because the feature columns are not independent (one gene across three conditions), the closed-form MP edge is a lower bound, so the number of signal directions was set empirically — the 95th percentile of the top eigenvalue over 100 column-permuted correlations ($\lambda_+ = 2.95$) leaves 92 signal directions. Communities were detected with Leiden (RBConfiguration objective) swept over resolution and 50 seeds (500 partitions), aggregated into a consensus co-assignment and re-clustered; communities with mean intra-community co-assignment s_c ≥ 0.8 are stability-gated. Consensus modularity was compared to a label-permutation null (z = 259). Community identity was tested by hypergeometric CORUM enrichment (BH-FDR) over complexes with ≥ 2 panel members.

## Disease-risk enrichment of communities

Autoimmune genetic-risk load per community was the summed per-gene maximum Open Targets `genetic_association` score across five diseases (SLE, RA, MS, Sjögren, T1D; score ≥ 0.01). Enrichment was tested against a null resampling 10,000 length-decile-matched sets from the 3,106 panel regulators (never the genome), reported with and without the chr6 MHC block (~28–34 Mb) excluded from module and universe. Results are reported as nominations, not causal claims.

## RPE1 third-cell-type concordance

The cross-cell-type concordance pipeline was applied unchanged to Replogle's essential-scale RPE1 bulk product (`rpe1_normalized_bulk_01.h5ad`, figshare plus 20029387), with per-gene Ensembl intersection and gene-level collapse identical to K562. A pre-registered overlap gate (proceed at ≥ 100 shared regulators; 962 shared) was checked before analysis. SAGA universality was read from the SAGA subunits present in the essential-scale set; the T-cell-specific side was assessed at the class level over the regulators classified T-specific in the CD4↔K562 comparison and also present in RPE1.

### Inductive prediction of unseen regulators

Leave-regulator-out prediction held out whole regulators (all three conditions at once) under 5-fold cross-validation and predicted their 6,000-dimensional trans-response from regulator side-features — transcription-factor and DNA-binding-domain annotation, STRING network degree, and GO/Pfam composition (605 features; 98–99 % panel coverage). Per-regulator held-out $R^2$ was measured against the training mean, so a mean predictor scores 0. A ridge model (a full-rank inductive matrix completion) and a CatBoost gradient-boosted learner (one model per component of a train-fold SVD of the response, $K \in \{10, 30\}$) were each compared against an identical model trained on row-permuted features. Real and permuted features performed identically out of sample, so the negative rests on the shuffled control alone.

### Code availability

All analysis code is available in the accompanying repository. The pipeline is organized by analysis: operator tensor construction, SVD/CP decomposition, and out-of-panel completion (`build_operator_tensor.py`, `decompose_operator_cp.py`, `operator_completion.py`, `operator_completion_stratified.py`), with numerical kernels in `_opkernels.py` and unit tests in `tests/test_opkernels.py`; spectral denoising, consensus community detection, and community identity (`operator_communities.py`, `operator_community_null.py`, `operator_community_identity.py`); the Section 3 insignia figure (`operator_insignia_figure.py`); community disease-risk enrichment (`community_gwas.py`); cross-cell-type concordance for K562 and RPE1 (`analyze_k562_concordance.py`, `analyze_rpe1_concordance.py`); and inductive prediction of unseen regulators (`imc_inductive.py`, `imc_nonlinear.py`). The full operator pipeline is driven by `make operator`; random seeds are fixed throughout, and all cited result tables are regenerated to `docs/tables/`.

## References


1. Dixit, A. *et al.* Perturb-seq: Dissecting molecular circuits with scalable single-cell RNA profiling of pooled genetic screens. *Cell* **167**, 1853–1866.e17 (2016).
2. Replogle, J. M. *et al.* Mapping information-rich genotype-phenotype landscapes with genome-scale perturb-seq. *Cell* **185**, 2559–2575.e28 (2022).
3. Zhu, R. *et al.* Genome-scale perturb-seq in primary human CD4+ t cells maps context-specific regulators of t cell programs and human immune traits. *bioRxiv* https://doi.org/10.64898/2025.12.23.696273 (2025) doi:10.64898/2025.12.23.696273.
4. Barry, T., Mason, K., Roeder, K. & Katsevich, E. Robust differential expression testing for single-cell CRISPR screens at low multiplicity of infection. *Genome Biology* **25**, (2024).
5. Peidli, S. *et al.* scPerturb: Harmonized single-cell perturbation data. *Nature Methods* **21**, 531–540 (2024).
6. Zhou, Y., Luo, K., Liang, L., Chen, M. & He, X. A new bayesian factor analysis method improves detection of genes and biological processes affected by perturbations in single-cell CRISPR screening. *Nature Methods* **20**, 1693–1703 (2023).
7. Ahlmann-Eltze, C., Huber, W. & Anders, S. Deep-learning-based gene perturbation effect prediction does not yet outperform simple linear baselines. *Nature Methods* **22**, 1657–1661 (2025).
8. Wei, Z. *et al.* Benchmarking algorithms for generalizable single-cell perturbation response prediction. *Nature Methods* **23**, 451–464 (2025).

9. Chevalley, M., Roohani, Y. H., Mehrjou, A., Leskovec, J. & Schwab, P. A large-scale benchmark for network inference from single-cell perturbation data. *Communications Biology* **8**, (2025).